Traffic Adaptive Network Guidance & Optimization (TANGO)

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Abstract

This project aimed to build a two-model system to optimize traffic by reducing the total amount of time spent waiting at intersections, and help predict traffic change as irregular conditions such as lane closure or opening of new transit lines. Only the first layer (ASCE), which reduces the time spent waiting at intersections, could be completed within the timeframe, and the transit planner layer (PIRA) could not be finished. ASCE has been proven to function and outperform the Max-Pressure algorithm under all four scenarios tested, with the largest gains on out-of-distribution conditions. Across 5 random seeds, the reductions in person-time-loss compared to Max-Pressure are as follows: AM Peak: -10.1%; PM Peak: -6.3%; Demand Surge: -11.0%; Midday Multimodal: -13.8%. ASCE additionally outperforms two further baselines on every scenario—SUMO's auto-generated Actuated Default signal programs and a naive Fixed-Time controller—while improving vehicle delay fairness (Jain index) by $\sim 30\%$ relative to Max-Pressure. The project code and trained models are publicly available.[1]

1 Introduction

Urban traffic congestion costs Toronto an estimated $6 billion annually in lost productivity [**?**]. This project was intended to be a two-model traffic optimization system for a corridor of 12 to 15 intersections in Toronto, specifically aiming to reduce total delay (first layer, Adaptive Signal Control Engine or "ASCE") and help predict traffic change due to a change in the infrastructure (second layer, or Planning Infrastructure Response Analyzer "PIRA").

ASCE proved to be a success—however, building the PIRA layer, which depended on the completion of the ASCE layer, proved to be ambitious within the given timeframe. This report will focus on the design and the results of the ASCE.

There are two baseline signal cycling systems which ASCE aimed to beat: Fixed-Time and Max-Pressure. Fixed-Time is simply letting intersections run blindly on a predefined timer. This is the most basic (and cheapest) system that should be easy to beat with any kind of adaptive signalling. Max-Pressure [**?**] is a decentralized (i.e. each intersection has no access to global traffic data) heuristic-based algorithm that prioritizes relieving the road with higher traffic. While this model performs well under regular traffic conditions it was trained on, it fails to adapt under irregular conditions (e.g. lane closure, sports event, subway disruptions).

ASCE is a decentralized model, where each intersection acts as an independent agent. It takes the current state at each signal intersection, including but not limited to queue lengths, incoming arrivals, approach speeds, and time of day. It then chooses a safe next signal decision stating the next phase to be served and duration of the green signal. This model aims to reduce delay and prevent queues from backing into other streets, even under irregular conditions.

[1] `https://github.com/yashnkapadia/ece324-TANGO`

Submitted to 38th Conference on Neural Information Processing Systems (NeurIPS 2024). Do not distribute.

The second layer, PIRA (Planning Infrastructure Response Analyzer), was meant to model the effect of construction, lane closures, and new public transit lines on traffic. For each scenario, the model would estimate how demand and road capacity will change and then recommend practical updates to signal timing and phasing, using insight from the first layer. This model aims to solve the problem of turning proposed projects into data-driven signal changes that keep traffic moving smoothly during and after the disruption. Following the proposal, ASCE's success criterion is a $\geq$10% reduction in person-time-loss relative to the Max-Pressure baseline (Section 2); PIRA was originally targeted to reach within the SUMO seed-noise floor (mean absolute percentage error $\leq$10%) but was not evaluated as it is incomplete.

2 Conditions of Success

For ASCE, the proposal criterion is that traffic engineering typically treats a 10% delay reduction as practically meaningful:

$$\mathrm{PTL}_{\mathrm{MAPPO}} \leq 0.90 \times \mathrm{PTL}_{\text{Max-Pressure}}, \tag{1}$$

where PTL denotes person-time-loss: the sum of per-vehicle excess travel time over free-flow conditions (SUMO `timeLoss`), weighted by estimated person occupancy (1.3 per car, 30 per transit vehicle). Evaluation uses 5 random seeds per scenario.

For PIRA, the target is a mean absolute percentage error of at most 10% across seeds, with inference under 5 seconds per scenario. As PIRA training is incomplete, this criterion is not evaluated in this report.

3 Data Pipeline and Simulation

This project uses two layers of data. The first layer is the City of Toronto Traffic Volumes Multimodal Intersection Turning Movement Counts (TMC) dataset [**?**], which records motor vehicle, bicycle, and pedestrian volumes at intersections in 15-minute intervals. The raw TMC dataset contains approximately 1 million samples dating back to 1980. Each record includes the intersection location, date, time period, approach direction, turning movement, and vehicle count. The second layer is a large simulated training dataset built on top of a real OpenStreetMap road network using the SUMO microsimulation platform [**?**], calibrated against the TMC volumes to produce 1.94 million reinforcement learning training samples.

3.1 Study Area

This project focuses on signalized intersections along Dundas Street West in Toronto, Ontario, Canada, from the intersection at University Avenue to the intersection at Bathurst Street. The signalized intersections selected on this corridor are shown in Figure 1.

This corridor was chosen because TMC data are available for it, it includes a streetcar route that is relevant for transit scenarios in the next phase of the project, and it is small enough (5 to 20 intersections) to iterate quickly while being large enough to capture signal coordination effects.

To support the simulated training dataset, the study area is first converted into a SUMO road network and then checked and corrected for signal control. The road network is created with the SUMO OSM Web Wizard [**?**], which pulls a bounding box from OpenStreetMap and produces a SUMO network file (`osm.net.xml`) that includes road segments (edges), intersections (junctions), lanes, connections, and traffic light programs. The bounding box covers roughly 200 meters north and south of Dundas Street between Bathurst and University. This network file is the first input to the MAPPO algorithm for training. After export, a mapping script links each known intersection to its corresponding SUMO junction by projecting geographic coordinates with `sumolib` and matching to the nearest junction ID. The result is an intersection map CSV that connects real intersection names to SUMO junction identifiers.

3.2 Custom Demand Generation Interface (Demand Studio)

After the network is prepared and mapped to real intersections, the next practical problem is turning observed counts into scenario files that can be run repeatedly for controller experiments. The custom

sumo_scenario_signalized_intersections.png

Figure 1: Generated SUMO scenario showing project area focus. The selected area is inspected using `sumolib` and annotated using `matplotlib` to highlight signalized intersections. The red markers indicate the locations of signalized intersections along the corridor.

Demand Studio interface was built for that step. It is a local Dash application that links a SUMO network file, visualizes the junction graph, maps TMC locations to nearby junctions, and generates new demand and scenario outputs from the parsed TMC dataset in one workflow. It was created to remove the manual loop of switching between separate SUMO tools, editing XML files by hand, and tracking experiment settings in separate notes. It also keeps the baseline files untouched by writing outputs to dedicated generated folders, which makes scenario iteration safer and easier to manage. For MAPPO work, this gives a single place to define the traffic scenario and the controller configuration that should be associated with that scenario.

Demand Studio combines demand creation, controller setup, and validation in one workflow from observed TMC counts to simulation-ready scenarios. In Demand settings, users select mapped TMC locations, set the maximum TMC-to-junction match distance, choose date policy and time window, define simulation begin and end, and include cars, trucks, buses, streetcars, and pedestrians with global and mode-specific scaling plus a bus-to-streetcar share. The app aggregates counts by approach and turning movement, converts them into SUMO `flow` and `personFlow` entries, and can enforce strict shortest-path checks to avoid invalid routes, which supports controlled construction of peak, off-peak, directional, and multimodal stress scenarios for MAPPO. In Signal and MAPPO settings, users choose MAPPO custom, baseline fixed time, or baseline max pressure, where fixed time generates a dedicated TLS additional file from EW and NS green, yellow, all-red, and offset settings with optional cluster-signal inclusion, while non fixed-time modes use the default TLS override file when available. Max-pressure parameters and MAPPO hyperparameters are saved in scenario metadata so demand design and controller assumptions stay coupled for reproducible experiments, even though training execution is outside this interface. Advanced settings improve transferability and

quality control through tunable heading ranges for N, S, E, and W approach classification on irregular network geometry, editable vehicle dynamics for cars, trucks, buses, and streetcars without manual route XML edits, and an optional short SUMO validation run that catches route and configuration issues early.

Further details about the custom Demand Studio web interface, UI walkthrough, and usage are given in the project documentation.[2]

3.3 Scenarios for ASCE Training

We used four scenarios for ASCE training: `am_peak`, `midday_multimodal`, `pm_peak`, and `demand_surge`.

The `am_peak` scenario is the reference case for ASCE training, meant to represent a relatively steady morning corridor where Max-Pressure is expected to perform well. Technically, it uses the latest parsed TMC data on the eight Dundas corridor intersections, with a morning window starting at minute 420 for 120 minutes, strict route checking enabled, and all five demand modes active with unit scaling. No artificial surge is added in Layer 2, while streetcars are still injected in both directions at a 360-second headway to keep regular transit pressure in the system. The run is configured with `simulation_seconds` 900 and `simulation_end` 1200, producing 174 vehicle flows and 8 pedestrian flows, with totals of 13,144 cars, 388 trucks, 168 buses, and 2,096 pedestrians.

The `midday_multimodal` scenario is the high-conflict daytime case, designed to stress multimodal interactions rather than pure commuter car dominance. In plain terms, it captures the period where pedestrian and transit activity is strongest, including the documented Spadina pedestrian intensity in the rationale. In settings terms, it uses the latest TMC records with a midday window starting at minute 660 for 180 minutes, all five modes enabled, unit mode scales, no synthetic surge, and the same bidirectional 360-second streetcar injection pattern. It runs with `simulation_seconds` 1200 and `simulation_end` 1500 and yields 177 vehicle flows and 8 pedestrian flows, with high totals across modes, including 20,928 cars, 679 trucks, 179 buses, and 6,538 pedestrians.

The `pm_peak` scenario represents the evening directional shift case, where demand pressure moves eastbound and pedestrian activity remains heavy. This scenario is configured from the latest TMC data using a PM window that starts at minute 960 and lasts 120 minutes, again with strict route checks, all five modes included, and no Layer 2 surge added. Transit pressure is kept comparable to the other base scenarios through 360-second eastbound and westbound streetcar injections. The configuration uses `simulation_seconds` 900 and `simulation_end` 1200 and generates 156 vehicle flows and 8 pedestrian flows, with totals of 18,810 cars, 239 trucks, 114 buses, and 5,347 pedestrians.

The `demand_surge` scenario is the event dispersal case, built to test how ASCE handles a short, strong directional shock on top of PM conditions. It keeps the same PM base demand setup as `pm_peak`, including latest TMC selection, `time_window_start` 960, `time_window_duration_min` 120, full multimodal inclusion, and 360-second bidirectional streetcar injections, then adds a Layer 2 eastbound surge with multiplier 2.0 from second 300 to second 600. This isolates a temporary directional spike without changing the rest of the scenario structure. It runs longer, with `simulation_seconds` 1200 and `simulation_end` 1500, and produces 192 vehicle flows and 8 pedestrian flows, with totals of 19,630 cars, 252 trucks, 124 buses, and 5,347 pedestrians.

The ASCE section below describes how the four scenarios were used for MAPPO training.

4 Modeling

4.1 Architecture: Action-Gate Residual MAPPO

We adopt Multi-Agent Proximal Policy Optimization (MAPPO) [**?**] with parameter sharing, following the Centralized Training with Decentralized Execution (CTDE) paradigm as used in the CityLight framework [**?**]. Each of the 12 signalized intersections acts as an independent agent sharing a common policy network. A centralized critic, used only during training, receives the concatenated observations from all agents to estimate state value via Generalized Advantage Estimation [**?**].

[2] `https://github.com/yashnkapadia/ece324-TANGO/blob/main/DATA.md`

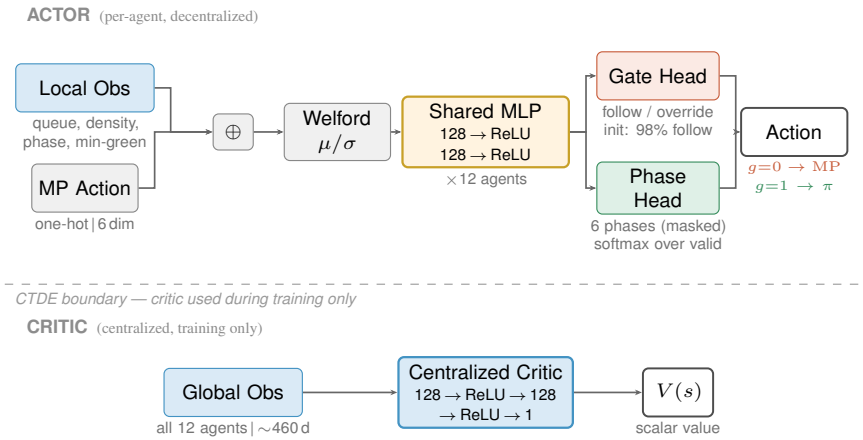


Figure 2: Action-gate residual MAPPO architecture. The actor (top) processes local observations augmented with the Max-Pressure recommendation and produces a gate decision (follow or override) and a phase selection. The centralized critic (bottom) is used during training only. All 12 intersection agents share the same actor and critic parameters.

For faster convergence and improved stability, the key architectural contribution is the **action-gate residual** design (Figure 2). At each time step, each agent receives a local observation concatenated with the Max-Pressure recommendation encoded as a one-hot vector. The shared actor network – a 2-layer ReLU MLP with 128 hidden units – produces two output heads:

- **Gate head** (2 outputs, softmax): a Bernoulli choice between *follow Max-Pressure* and *override with learned phase*.
- **Phase head** (K outputs, softmax): a distribution over legal signal phases, masked to valid phases per intersection ($K \in \{2, \ldots, 6\}$).

The gate head is initialized with a bias of $[-2.0, +2.0]$, corresponding to an initial probability of approximately 98% to follow Max-Pressure. This ensures the policy begins at Max-Pressure performance and focuses learning on situations where the heuristic is suboptimal. The action log-probability used for PPO updates is the joint log-probability of the gate decision and (if overriding) the phase selection, so both heads are trained end-to-end via a single PPO loss.

Observations. Each agent observes local traffic state: current phase, per-lane density and queue fractions, and the Max-Pressure recommended phase as a one-hot vector. Feature dimensions vary by intersection topology (15 to 49 after padding); full details appear in Appendix E. Features are normalized online using Welford's running mean and variance [**?**]. The centralized critic receives all 12 agents' observations concatenated ($\sim$460 dimensions) and is discarded at deployment.

4.2 Reward Design

The reward function underwent three iterations (detailed in Appendix D). The interim report's multi-objective reward—combining delay, throughput, and global fairness—left MAPPO 25% behind Max-Pressure because the throughput term dominated the delay gradient. An intermediate attempt to normalize person-weighted throughput into the reward amplified the problem further.

The final reward, `person_objective`, removes throughput entirely and instead makes the delay signal richer:

$$r_i = -\log\left(1 + \frac{\sum_v w_v \cdot t_v}{\sum_v w_v}\right) + 0.25 \cdot J\left(D_{\mathrm{NS}}^{(i)},\, D_{\mathrm{EW}}^{(i)}\right), \tag{2}$$

where the sum is over vehicles v on edges approaching intersection i, w_v is the person occupancy weight (1.3 for cars, 30.0 for transit), t_v is the waiting time, $D_{\mathrm{NS}}^{(i)}$ and $D_{\mathrm{EW}}^{(i)}$ are person-weighted delays on the north–south and east–west approaches, and $J(\cdot)$ is the two-element Jain index. Dividing person-delay by person-throughput yields a per-person delay *rate* that scales naturally with traffic

volume. The local directional Jain term is directly actionable per agent, unlike the global throughput fairness used at interim. Person weighting aligns the reward with the evaluation KPI.

4.3 Training Pipeline

Training proceeds in two phases, both using the `person_objective` reward.

Phase 1: Warm-start (200 episodes, AM Peak only). The policy is trained from scratch on the AM Peak scenario to establish basic signal coordination before encountering harder scenarios. This phase uses a single SUMO worker and base learning rates (actor: 3×10^{-4}, critic: 1×10^{-3}). By episode 200, MAPPO achieves a person-time-loss ratio of approximately 0.90 relative to Max-Pressure on the AM Peak scenario.

Phase 2: Curriculum training (600 episodes, 4 scenarios). Starting from the warm-start checkpoint, the policy trains on all four scenarios in round-robin order (Table **??**), using 8 parallel SUMO workers via Python multiprocessing with `libsumo`. Learning rates are scaled by $1/\sqrt{8}$ to compensate for the larger effective batch size from parallel collection [**?**], yielding actor 1.06×10^{-4} and critic 3.54×10^{-4}. Episodes run for 900 s (AM/PM Peak) or 1200 s (Demand Surge, Midday Multimodal) at 5 s decision intervals.

A best-model checkpoint is saved whenever inline evaluation (run every 48 episodes) achieves a new minimum worst-case MAPPO/MP ratio across all four scenarios. The best checkpoint occurs at curriculum episode 432 (total episode 632), after which training regresses due to conflicting gradient directions across scenarios—a known challenge in round-robin curriculum training. The final model (episode 600) performs measurably worse than the best checkpoint (Table 1), confirming that early stopping on worst-case evaluation is essential for curriculum MAPPO.

Infrastructure. The SUMO-RL environment interface [**?**] was vendored and modified to preserve native NEMA-type traffic-light programs from the Toronto network. The upstream library replaces actuated signal programs with simplified static programs, which caused simulation crashes at 4 of 12 intersections. The vendored version uses `setPhase()` to select among existing program phases rather than constructing new signal states. Full hyperparameters appear in Appendix A.

4.4 Results

Table 1 reports person-time-loss (mean $\pm$ standard deviation over 5 seeds) for the best curriculum checkpoint across all four evaluation scenarios. Three baseline controllers are compared: Max-Pressure (reactive, selects phase with highest queue differential), Actuated Default (SUMO's auto-generated actuated programs from the OSM network import, which dynamically extend green phases when vehicles are detected), and Fixed-Time (uniform 30 s green cycle). Baselines are ordered by person time-loss (PTL) within each scenario. MAPPO achieves the lowest person-time-loss on all four scenarios.

The best curriculum checkpoint meets the 10% improvement target on AM Peak, Demand Surge, and Midday Multimodal traffic. PM Peak, the scenario with the highest total volume and a tidal reversal pattern, shows a 6.3% improvement over Max-Pressure. We hypothesize that the relative under-performance of MAPPO in PM Peak is due to the complexity of managing high-volume, reversed flow traffic conditions and expect it to be ameliorated with further training. The relative ordering of baselines shifts across scenarios: Max-Pressure is the strongest baseline on AM Peak and Midday Multimodal, while Actuated Default is strongest on PM Peak and Demand Surge—the two high-volume arterial-dominant scenarios where its built-in phase extension provides an advantage similar to Max-Pressure's queue responsiveness. MAPPO beats all baselines on every scenario regardless of which baseline is strongest.

Secondary metrics. MAPPO also improves on secondary metrics relative to Max-Pressure (full table in Appendix F). Throughput increases by 9–16% (more vehicles complete trips). Vehicle delay fairness (Jain index) improves from 0.34–0.40 (MP) to 0.43–0.52 (MAPPO)—a relative improvement of $\sim$30%, though no controller reaches the proposal's aspirational target of 0.8. The low absolute Jain values reflect the inherent heterogeneity of a real corridor where some approaches consistently experience more delay than others.

Table 1: Person-time-loss (seconds, mean ± std over 5 seeds) across four evaluation scenarios. Three baselines are compared: Max-Pressure (reactive, queue-differential), Actuated Default (SUMO's auto-generated actuated programs from OSM), and Fixed-Time (uniform 30 s green cycle). MAPPO/MP values below 0.90 meet the proposal target (bold).

Scenario	Controller	PTL (s)	MAPPO/MP	≥10%?
AM Peak	Max-Pressure	427,943 ± 12,390	—	
	Actuated Default	439,932 ± 15,442	—	
	Fixed-Time	456,689 ± 16,047	—	
	MAPPO (best)	384,598 ± 5,946	**0.899**	✓
PM Peak	Actuated Default	490,484 ± 9,981	—	
	Fixed-Time	492,550 ± 15,828	—	
	Max-Pressure	518,755 ± 15,072	—	
	MAPPO (best)	486,226 ± 28,139	0.937	
Demand Surge	Actuated Default	453,349 ± 9,555	—	
	Fixed-Time	462,132 ± 7,210	—	
	Max-Pressure	503,202 ± 29,971	—	
	MAPPO (best)	447,626 ± 7,284	**0.890**	✓
Midday Mult.	Fixed-Time	500,876 ± 4,888	—	
	Actuated Default	508,168 ± 29,727	—	
	Max-Pressure	544,741 ± 18,160	—	
	MAPPO (best)	469,303 ± 16,586	**0.862**	✓

Comparison with the final model. The final curriculum model (episode 600) achieves MAPPO/MP ratios of 0.929 (AM), 0.926 (PM), 0.930 (Surge), and 0.882 (Midday)—uniformly worse than the best checkpoint. This regression after episode 432 reflects gradient interference from the round-robin curriculum: updates that improve one scenario's objective can degrade another's. Best-model checkpointing is therefore essential.

Gate behavior. The gate fraction (proportion of time steps where the agent overrides Max-Pressure) starts near 2% due to the initialization bias and rises to approximately 30% by the end of curriculum training (Figure 4b). This confirms that the policy learns to selectively override rather than replacing Max-Pressure wholesale. The override rate is highest during demand transitions (surge onset, tidal reversal) and lowest during stable steady-state flow—consistent with the hypothesis that Max-Pressure is near-optimal under stationary demand but suboptimal during transients.

4.5 Analysis

Why MAPPO wins. Max-Pressure selects the phase with the highest queue differential at each step, which is locally optimal for throughput but ignores three factors that matter for person-time-loss: (1) person occupancy—a bus carrying 30 people stuck behind a queue costs 30× more than a single car; (2) temporal coordination—green waves that benefit upstream intersections may not emerge from independent greedy decisions; and (3) anticipation of demand transitions—Max-Pressure reacts to current queues but cannot anticipate a surge that will arrive in 30 seconds.

ASCE's advantage is largest on the Midday Multimodal scenario (13.8% improvement), where high pedestrian volumes at Spadina Avenue and frequent streetcar service create person-weighted delays that Max-Pressure's vehicle-count-based heuristic cannot prioritize. The Demand Surge scenario (11.0%) benefits from ASCE learning to preemptively redistribute green time before the 2× eastbound surge saturates downstream intersections.

Limitations. The training curve does not converge smoothly; round-robin curriculum causes oscillation as scenario-specific gradients conflict. The best-model checkpoint mitigates but does not solve this. More principled curriculum strategies (e.g., prioritized experience replay across scenarios, or multi-task loss weighting) may improve stability. PM Peak remains below the 10% target, suggesting that the current curriculum does not allocate sufficient training to the highest-volume scenario. The Fixed-Time baseline uses a uniform 30 s green cycle rather than the actual Toronto signal timing programs; a more realistic comparison would use the real fixed-time plans

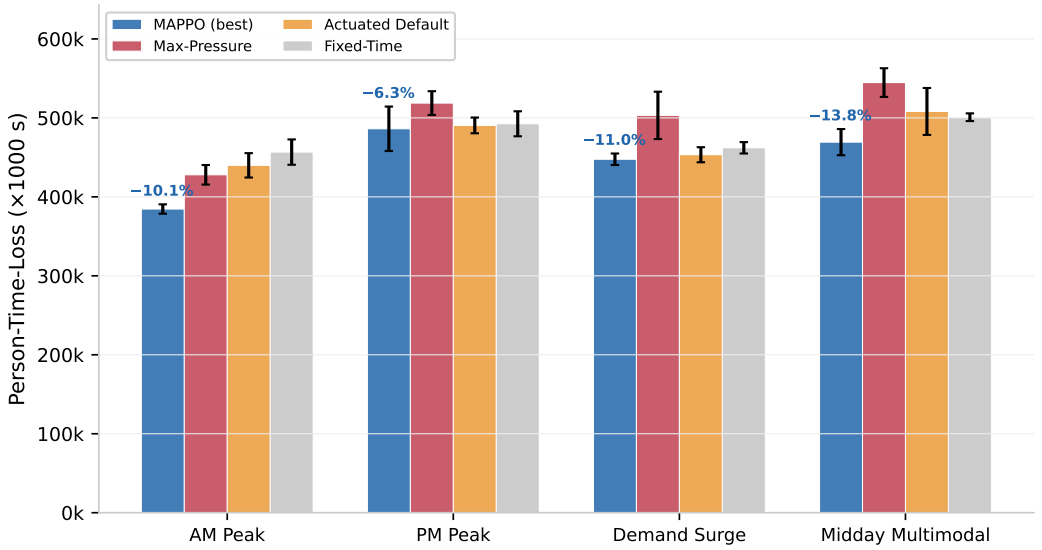
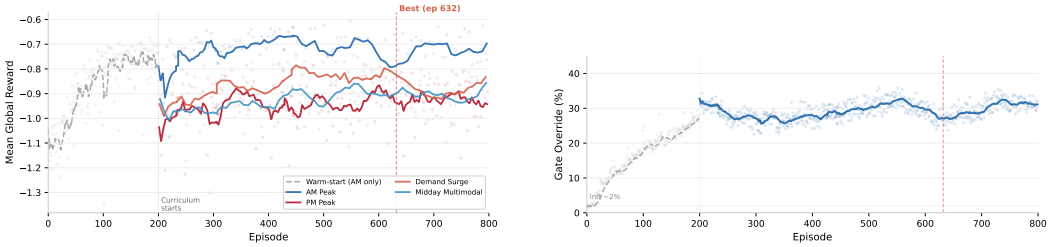


Figure 3: Person-time-loss by scenario and controller for the best curriculum checkpoint. Error bars show ± 1 standard deviation over 5 seeds. Percentage labels indicate MAPPO improvement relative to Max-Pressure.



(a) Mean global reward per episode.

(b) Gate override fraction.

Figure 4: ASCE training dynamics. (a) Reward per episode colored by scenario; the first 200 episodes are warm-start on AM Peak only, followed by round-robin curriculum. The dashed line marks the best checkpoint at episode 632. (b) Fraction of time steps where the agent overrides Max-Pressure, rising from $\sim$2% (initialization bias) to $\sim$30%.

from the City. Additionally, legally mandated safety constraints (minimum green, maximum green, pedestrian recall) are not enforced on MAPPO's phase selections, which would be required for any real-world deployment.

4.6 PIRA: Scenario Planning Surrogate (Incomplete)

Although incomplete, some progress was made on the PIRA layer. The general pipeline was established, having a Python file (`pira.py`) that takes a Parquet rollout file from ASCE as well as the OSM file used for SUMO, converts the road network into a graph neural network operating over intersections (nodes) and road segments (edges), and exports a trained model which can then be used in a separate Python file dedicated to planning road infrastructure changes (`planner.py`). Both files live under `ece324_tango/pira/`, and dataset preparation is handled by `scripts/generate_baseline_dataset.py` (Max-Pressure and Fixed-Time rollouts) and the companion `scripts/generate_asce_dataset.py` (trained-policy rollouts), which together produce the Parquet files `pira.py` consumes. However, the model was not trained on real data—it was trained on synthetic data to show that the pipeline works as intended, and so was the planner. The planner used in the presentation GIF was also running on garbage data; it only served as a communication tool for the audience for them to understand what we wished to achieve. PIRA's

accuracy and inference-time targets (MAPE $\leq 10\%$, <5 s) are therefore not evaluated, and completing real-data integration is the primary remaining work item.

5 Future Actions

As of this report, the following two actions are still required to achieve what the project originally proposed to achieve.

Firstly, PIRA has not been completed yet. This is the most significant part that requires future work.

Secondly, although ASCE has been tested to work and beat both baseline models under various conditions, this was not done on real traffic. While real traffic statistics were used, much of the model was trained on SUMO simulation output, taking this as the ground truth. While there is no realistic way to iterate on this in real life, this is a potential improvement to consider.

6 Potential Impacts

This project has the potential to reduce total delays at intersections, reducing the productivity loss. However, there is also a potential that reduced traffic simply induces more car demand, making the city worse off from encouraged car usage and lowered interest in public transit investments.

7 Reproducibility

All code, trained models, and evaluation scripts are available at `https://github.com/yashnkapadia/ece324-TANGO`. The repository includes:

- `pixi.toml` — fully locked dependency specification (CUDA 12.6, PyTorch 2.7.1, SUMO 1.26.0)
- `pixi run train-asce-toronto-demand` — reproduces the training pipeline
- `pixi run eval-asce-toronto-demand` — reproduces evaluation
- `models/asce_mappo_curriculum_best.pt` — best curriculum checkpoint
- `sumo/demand/curriculum/` — all four scenario demand files

Training on an NVIDIA RTX 4070 Laptop GPU takes approximately 8 hours (200 warm-start episodes + 600 curriculum episodes with 8 parallel workers). Total compute for all reported experiments, including preliminary and failed runs, is approximately 40 GPU-hours.

Statement of Scope Changes from Proposal

The following changes were made relative to the original proposal:

1. **PIRA training not completed.** The GNN architecture, training loop, and synthetic dataset generator are implemented and run end-to-end, but training on the real ASCE-generated baseline dataset converged poorly within the project timeline due to feature-engineering issues mapping ASCE rollouts into the GNN's expected graph input format. The scenario planning capability described in the proposal is therefore not evaluated.
2. **IPPO baseline replaced with action-gate ablation.** The proposal listed Independent PPO as a baseline to ablate the centralized critic. We instead focused on the action-gate residual architecture, which provides a more informative ablation of the Max-Pressure initialization benefit. The centralized critic ablation remains future work.
3. **Stretch goals not attempted.** CityLight-style neighborhood rewards, IG-RL transfer, CoLight graph attention, and DSA-QMIX were not implemented. Time was spent instead on reward engineering and the action-gate architecture, which proved sufficient to meet the primary success criterion.
4. **Actuated Default baseline added.** SUMO's auto-generated actuated signal programs from the OSM network were added as a third baseline (alongside Max-Pressure and Fixed-Time), enabled by the vendored `sumo-rl` modifications that preserve native traffic-light programs.
5. **Evaluation expanded to four scenarios.** The proposal described evaluation on a single scenario; the final evaluation covers four distinct demand patterns (AM Peak, PM Peak,

Demand Surge, Midday Multimodal) with multimodal traffic including pedestrians and TTC streetcars, generated through the custom Demand Studio interface (Section 3.3).

A MAPPO Hyperparameters

Parameter	Value
Hidden dimension	128
Hidden layers	2 (actor and critic)
Activation	ReLU
Actor learning rate (base)	3×10^{-4}
Critic learning rate (base)	1×10^{-3}
LR scaling (parallel)	$1/\sqrt{N_{\text{workers}}}$
Optimizer	Adam
Discount (γ)	0.99
GAE λ	0.95
PPO clip ϵ	0.2
Entropy coefficient	0.01
Value loss coefficient	0.5
Gradient clip (max norm)	0.5
PPO epochs per update	5
Minibatch size	512
Gate initialization bias	-2.0 (98% follow MP)
Observation normalization	Welford running mean/variance
Parallel workers	8 (libsumo)

B Person-Time-Loss Definition

Person-time-loss is defined as:

$$\text{PTL} = \sum_{v \in \mathcal{V}} \Delta t_v^{\text{loss}} \cdot w_v, \tag{3}$$

where $\mathcal{V}$ is the set of all vehicles active during the episode, Δt_v^{loss} is the SUMO-reported time-loss (excess travel time over free-flow speed) for vehicle v, and w_v is the person occupancy weight: 1.3 for cars and 30.0 for TTC buses, streetcars, and transit vehicles.

C Metric Definitions

Metric	Definition
Person-time-loss (s)	See Appendix B.
Vehicle delay fairness (Jain)	Jain's index [**?**] computed over per-vehicle final time-loss values for all vehicles that complete their trip during the episode.
Throughput	Number of vehicles that complete their trip within the episode.
Avg trip time (s)	Mean travel time from departure to arrival over all arrived vehicles.
Gate fraction	Proportion of agent-steps where the gate head selects "override" rather than "follow Max-Pressure."

D Reward Function Iterations

The final `person_objective` reward (Eq. 2) was arrived at through three iterations, each addressing a specific failure mode. Table 2 summarizes the progression.

Table 2: Reward function ablation on AM Peak. MAPPO/MP is the person-time-loss ratio (lower is better).

Reward mode	MAPPO/MP	Scenario	Failure mode
`time_loss`	>1.0	AM Peak	Too sparse; worse than Fixed-Time
`objective`	0.93	AM Peak	Throughput bias dominates gradient
person + throughput	≈1.0	AM Peak	Throughput amplified by normalization
`person_objective`	**0.90**	AM Peak	Selected (see Table 1)

Iteration 1: `objective` **(interim report).** Combined delay, throughput, and global fairness: $r_i = -\log(1 + d_i) + \log(1 + \tau_i) + 0.25 \cdot J(\boldsymbol{\tau})$, where d_i is waiting time at intersection i, τ_i is vehicle count, and $J(\boldsymbol{\tau})$ is Jain's index [**?**] over per-agent throughputs. This enabled MAPPO to beat Fixed-Time but left it 25% behind Max-Pressure. The throughput term $\log(1 + \tau_i)$ acted as a near-constant positive bias that dominated the delay gradient, causing the policy to optimize for flow rather than delay reduction. Evaluation results were highly variable across seeds.

Iteration 2: Person-weighted with throughput (abandoned). To fix the bias, we switched to person-weighted metrics (occupancy weighting: 1.3 for cars, 30 for transit) and attempted to normalize the throughput term. Normalizing throughput into the formula amplified its relative magnitude further; the model learned to maximize flow volume while ignoring delay entirely (MAPPO/MP ≈ 1.0).

Iteration 3: `person_objective` **(final).** The key insight was that throughput should not appear in the reward at all. It was originally added because the raw `time_loss` signal was too sparse (a lesson from the interim report). The correct fix was to make the delay signal richer by (1) computing a per-person delay rate instead of a raw accumulator, (2) using local NS/EW directional Jain fairness instead of global throughput fairness, and (3) person-weighting to align with the evaluation KPI. See Section 4.2 for the formula.

E Observation and Feature Dimensions

Each agent's observation is a variable-length vector whose dimension depends on the intersection's topology:

Table 3: Observation feature blocks per agent. Total dimension ranges from 9 to 43 (base) plus 6 (MP augmentation) = 15 to 49, padded to a common maximum.

Block	Size	Description
Phase one-hot	2–6	Current green phase encoding
Min-green flag	1	1 if phase may switch, else 0
Lane density	3–18	vehicles/capacity per incoming lane, $[0, 1]$
Lane queue	3–18	halted vehicles/capacity per lane, $[0, 1]$
MP one-hot	6	Max-Pressure recommended phase (gate mode)
Total	15–49	Padded to max across all 12 agents

The centralized critic receives the concatenation of all 12 agents' normalized observations ($\sim$460 dimensions). Observation normalization uses Welford's online algorithm [**?**] to maintain per-feature running mean and variance, applied separately to local (actor) and global (critic) observation streams.

The actor network (GatedActor) has 128×2 hidden units with ReLU activations. Its two output heads are:

- **Gate head**: Linear($128 \rightarrow 2$), softmax. Initialization bias $[-2.0, +2.0]$ yields $\sim$98% follow-MP at start.
- **Phase head**: Linear($128 \rightarrow 6$), softmax with invalid-action masking (logits set to $-\infty$ for illegal phases). Valid phases range from 2 to 6 per intersection.

F Full Evaluation Metrics

Table 4 reports all evaluation metrics for the best curriculum checkpoint across scenarios (mean ± std, 5 seeds).

Table 4: Complete evaluation metrics for `curriculum_best` checkpoint.

Scenario	Controller	Avg Trip (s)	Arrived	Jain	Gate %
AM Peak	Max-Pressure	76.5 ± 5.4	631 ± 31	0.340 ± 0.019	—
	Fixed-Time	75.1 ± 1.9	664 ± 14	0.307 ± 0.016	—
	MAPPO	73.4 ± 2.8	688 ± 31	0.432 ± 0.024	26.7
PM Peak	Max-Pressure	102.7 ± 4.9	542 ± 19	0.404 ± 0.013	—
	Fixed-Time	85.6 ± 3.3	687 ± 18	0.361 ± 0.031	—
	MAPPO	102.3 ± 5.6	604 ± 39	0.506 ± 0.050	27.4
Demand Surge	Max-Pressure	93.4 ± 1.8	539 ± 87	0.389 ± 0.018	—
	Fixed-Time	81.6 ± 2.2	702 ± 14	0.355 ± 0.007	—
	MAPPO	94.7 ± 0.6	627 ± 31	0.521 ± 0.030	27.3
Midday Mult.	Max-Pressure	93.4 ± 3.5	558 ± 32	0.369 ± 0.040	—
	Fixed-Time	85.5 ± 2.9	605 ± 26	0.317 ± 0.014	—
	MAPPO	92.3 ± 6.0	611 ± 15	0.477 ± 0.021	30.2

MAPPO achieves the lowest average trip time on AM Peak and is comparable to Max-Pressure on other scenarios. Fixed-Time achieves lower trip times on PM Peak and Demand Surge because it cycles uniformly and clears more vehicles through the network, but at the cost of substantially worse person-weighted delay and fairness. MAPPO's Jain index is 27–34% higher than Max-Pressure across all scenarios, indicating more equitable delay distribution, though no controller approaches the proposal target of 0.8.

G Summary of Changes from Interim Report

Since the interim report, several modifications have been made.

Firstly, the ASCE algorithm was upgraded to be action-gated with Max-Pressure as the base instead of a simple MAPPO algorithm. This was one of the main future works discussed in the interim report, as the intended benefit of using ASCE over Max-Pressure is that it outperforms Max-Pressure under irregular conditions, while performing at least as well as Max-Pressure under regular conditions. In the interim report, Max-Pressure used to outperform ASCE in every scenario.

Specifically, four different scenarios (AM peak, PM peak, midday multimodal—i.e. peak pedestrian traffic—and demand surge) were made based on real TMC corridor data for our chosen intersections, and the model was trained through curriculum training. Now, ASCE outperforms Max-Pressure under all conditions, regular or irregular. To accomplish curriculum training, batched reinforcement learning was also implemented.

This also means that the model now includes the effect of buses, pedestrians, and paralleling transit lines.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract states specific MAPPO/MP improvement percentages that are directly supported by Table 1. The incomplete status of PIRA is explicitly noted.

2. **Limitations**

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 4.5 discusses PM Peak underperformance, curriculum instability, and remaining engineering limitations. Section **??** lists scope reductions.

3. **Theory Assumptions and Proofs**

   Answer: [NA]

   Justification: No theoretical results are claimed.

4. **Experimental Result Reproducibility**

   Answer: [Yes]

   Justification: Full code, trained models, locked dependencies, and reproduction commands are provided (Section 7).

5. **Open access to data and code**

   Answer: [Yes]

   Justification: GitHub repository linked in Section 7 with all code, models, and data.

6. **Experimental Setting/Details**

   Answer: [Yes]

   Justification: Hyperparameters in Appendix A, training pipeline in Section 4.3, evaluation protocol in Section 4.4.

7. **Experiment Statistical Significance**

   Answer: [Yes]

   Justification: All results report mean $\pm$ std over 5 random seeds.

8. **Experiments Compute Resources**

   Answer: [Yes]

   Justification: GPU type, training time, and total compute budget reported in Section 7.

9. **Code Of Ethics**

   Answer: [Yes]

   Justification: Research uses only publicly available traffic statistics and open-source simulation tools.

10. **Broader Impacts**

    Answer: [Yes]

    Justification: Section 6 discusses both positive (delay reduction) and negative (induced demand) potential impacts.

11. **Safeguards**

    Answer: [NA]

    Justification: The project uses anonymized, aggregated traffic statistics with no privacy or misuse risk.

12. **Licenses for existing assets**

    Answer: [Yes]

    Justification: SUMO (EPL-2.0), SUMO-RL (MIT), MAPPO reference implementation (MIT) are cited.

13. **New Assets**

    Answer: [Yes]

    Justification: Trained models and vendored SUMO-RL modifications released under the project repository.

14. **Crowdsourcing and Research with Human Subjects**

    Answer: [NA]

15. **IRB Approvals**

    Answer: [NA]

Generative AI Usage Statement

Claude Code (Anthropic, Claude Opus) was used as a development assistant throughout this project for: code implementation and debugging of the ASCE training pipeline, reward function iteration, parallel training infrastructure, and drafting portions of this report. All generated code was reviewed, tested, and modified by the authors. Claude Code was not used to generate experimental results or evaluation metrics—all reported numbers come from SUMO simulation runs.